

Probability, Sampling and sampling distribution

Probability and Probability Distributions

Session Objectives

At the end of the session students should be able to:

- ✓ Understand the concepts and characteristics of probabilities and probability distributions.
- ✓ Compute probabilities of events
- ✓ Differentiate between the discrete and continuous distributions
- ✓ Understand the concepts and uses of the standard normal distribution

Introduction

- ▶ It should be noted that medical and public health practitioners seldom can predict an outcome with absolute certainty.
- ▶ E.g., to formulate a diagnosis, a physician must rely on available diagnostic information about a patient
 - History and physical examination
 - Laboratory studies, X-ray findings, etc
- ▶ Although no test result is absolutely accurate, it does affect the probability of the presence (or absence) of a disease.

Introduction...

- Probability is the language of chance.
- Probability is the chance of an outcome of an experiment.
- It is the measure of how likely an outcome is to occur.
- These numerical measures of uncertainties/certainties are called probabilities of occurrence of the events.
- The deliberate use of **chance is the central idea** of statistical designs for producing data.
- **Probabilities are used in everyday communication**

Introduction...

- ▶ An understanding of probability is fundamental for quantifying the uncertainty that is inherent in the decision-making process
- Probability theory also allows us to draw conclusions about a population of patients based on known information about a sample of patients drawn from that population
- Probability theory, in fact, is the foundation of statistical inferences.

Introduction...

A fair coin has been tossed 800 times.

Here is a record of the number of times it came up head, and the proportion of heads in the throws already made:

Number of tosses	Number of heads in the last 80 tosses	Cumulative number of heads	Proportion of heads in total number of tosses
80	38	38	.475
160	40	78	.488
240	47	125	.521
320	39	164	.513
400	40	204	.510
480	45	249	.519
560	32	281	.502
640	40	321	.502
720	38	359	.499
800	42	401	.501

As the result of this experiment, we say that the probability of heads **[notation: $\text{pr}(H)$]** on one toss of this coin is about .50. That is, $\text{Pr}(H) = 0.5$

Introduction...

- **Proportion measures size**; it is a descriptive statistic.
- **Probability measures chance**.

Example

- ✓ a box containing 100 marbles, 90 of them green and the other 10 blue.
- ✓ If the question is: “Are there green marbles in the box?”, someone who saw the box’s contents would answer “90%.”
- ✓ But if the question is: “If I take one marble at random, do you think I would have a green one?”, the answer would be “90% chance.”

Definition and Basic concepts

- **Random experiment:** Random experiment is one whose results depend on chance, that is the result cannot be predicted. Tossing of coins, throwing of dice are some examples of random experiments.
- **Trial:** Performing a random experiment is called a trial.
- **Outcomes:** The results of a random experiment are called its outcomes. When two coins are tossed the possible outcomes are HH, HT, TH, TT.

Definition ...

- **Event:** An outcome or a combination of outcomes of a random **experiment** is called an event. For example tossing of a coin is a random experiment and getting a head or tail is an event.
- **Sample space:** Each conceivable outcome of an experiment is called a sample point.
- The totality of all sample points is called a sample space and is denoted by S . For example, when a coin is tossed, the sample space is $S = \{ H, T \}$. H and T are the sample points of the sample space S .

Definition ...

- **Equally likely events:** Two or more events are said to be equally likely if each **one of them has an equal chance of occurring**. For example in tossing of a coin, the event of getting a head and the event of getting a tail are equally likely events.
- **Mutually exclusive events:** Two or more events are said to be mutually exclusive, when **the occurrence of any one event excludes the occurrence of the other event**.
- Mutually exclusive events cannot occur simultaneously.

Definition ...

- **Complementary events:** The event 'A occurs' and the event 'A does not occur' are called complementary events to each other.
- The event 'A does not occur' is denoted by A' or A^c .
- The event and its complements are mutually exclusive.
- For example in throwing a die, the event of getting odd numbers is $\{1, 3, 5\}$ and getting even numbers is $\{2, 4, 6\}$
- These two events are mutually exclusive and complement to each other.

Definition ...

- **Intersection(And B):** a set containing all elements in both A and B.
- **Union(Or):** a set containing all elements in A or B or both
- **Simple or elementary event:** If an event **E** consists of a single outcome, then it is called a simple or elementary event.
- **Impossible event:** This is an event which will never occur.
- **Certain event** is an event which is sure to occur.

Approaches to measuring probability

- Classical approach

Assigning probabilities based on the assumption of equally likely outcomes.

- Relative Frequency approach

Assigning probabilities based on experimentation or historical data.

- Subjective approach

Assigning probabilities based on the assignor's judgment.

- Axiomic approach

Probability...

The Classical Probability Concept:

- If there are n *equally likely possibilities*, of which one must occur and m are regarded as favourable, or as a “success,” then the probability of a “success” is m/n .

Example: What is the probability of rolling a 6 with a *well-balanced die*?

In this case, $m=1$ and $n=6$, so that the probability is $1/6$
 $= 0.167$

Probability...

- **Relative Frequency Approach:** Relative frequency of an event occurring in an infinitely large number of trials
- (proportion of times the outcome would occur if we observed the random process an infinite number of times.)
- Example: If records show that 60 out of 100,000 bulbs produced are defective. What is the probability of a newly produced bulb to be defective?
- Solution: Let A be the event that the newly produced bulb is defective

$$P(A) = \frac{60}{100,000} = 0.0006$$

sampling

Objectives

- At the end this session students are able to:
 - Describe methods of sampling
 - list assumptions of each sampling procedure
 - Apply appropriate sampling method for different scenarios

Sampling

- ✦ It is not easy to collect all the information about population and
- ✦ Also it is not possible to study the characteristics of the entire population (finite or infinite)
- ✦ Due to time factor, cost factor and other constraints.
- ✦ Thus we need sample.
- ✦ Sample is a finite subset of statistical individuals in a population and the number of individuals in a sample is called the sample size.

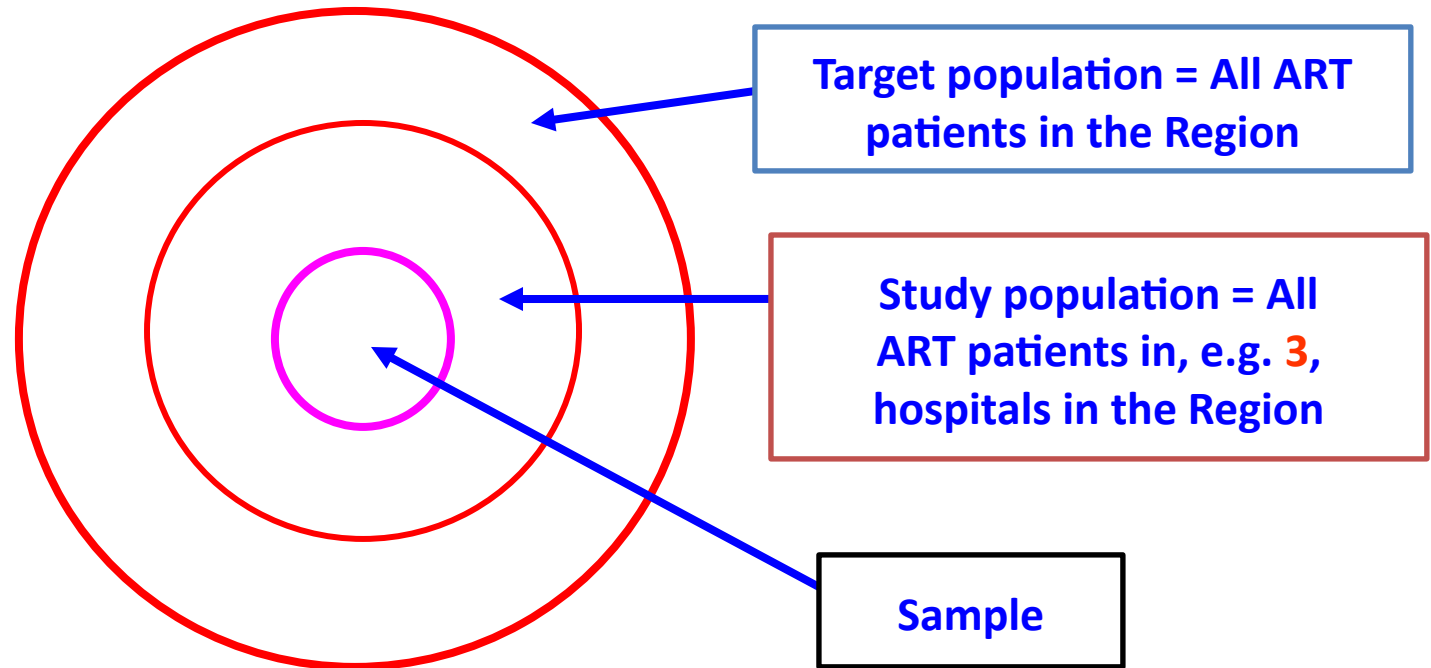
Common terms used in sampling

- **Population**: it is the collection of all items of interest.
- **Sampling**: It is the method by which we select a sample from the population
- *Reference population (or target population)*: the population of interest to whom the researchers would like to make generalizations.

Common terms ----

- *Sample population*: the actual group in which the study is conducted = **Sample**
- *Study population*: the subset of the target population from which a sample will be drawn.
- *Study unit*: the population on which information will be collected(measurement is done).
- *Sampling frame*: It is the list of all the sampling units in the source population and from which a random sample is to be drawn.

Researchers are interested to know about factors associated with ART use among HIV/AIDS patients attending certain hospitals in a given Region



Advantages of sampling:

- **Feasibility**: Sampling may be the only feasible method of collecting information.
- **Reduced cost**: Sampling reduces demands on resource such as finance, personnel, and material.
- **Greater accuracy**: Sampling may lead to better accuracy of collecting data
- **Greater speed**: Data can be collected and summarized more quickly.

Disadvantage

- There is always a sampling error
- Sampling may create a feeling of discrimination within the population
- Sampling may be inadvisable where every unit in the population is legally required to have a record

Errors in sampling

1. Sampling error/ Random error

- ▶ A sample is expected to mirror the population from which it comes,
- ▶ However, there is no guarantee that any sample will be precisely representative of the population.
- ▶ The **uncertainty** associated with an estimate that is based on data gathered from a sample of the population rather than the full population is **known as sampling error**.

Error sampling cont'd...

- ▶ Sampling errors are the **random variations** in the sample estimates around the true population parameters
- ▶ **No sample is the exact mirror image of the population**
 - Sampling error (chance)
 - ✓ Can not be avoided or totally eliminated
- ▶ Sampling error **decreases with the increase** in the size of the sample, and it happens to be of a smaller magnitude in case of homogeneous population.
- ▶ When $n = N \Rightarrow \text{sampling error} = 0$

Error in Sampling cont---

2. Non Sampling Error (Measurement Error)

- ▶ It is a type of systematic error in the design or conduct of a sampling procedure
- ▶ Which results in distortion of the sample, so that it is no longer representative of the reference population.
- ▶ We can eliminate or reduce the non-sampling error (bias) by careful design of the sampling procedure and not by increasing the sample size.
- ▶ It can occur whether the total study population or a sample is being used.

Error in sampling cont'd...

- ▶ It may either be produced by participants in the study or be an innocent by product of the **sampling plans and procedures**
- ▶ These biased observations can be innocent but very **devastating** to the findings of the study.
- ▶ **Example:** If you take male students only from a student dormitory in Ethiopia in order to determine the proportion of smokers, you would result in an overestimate, since females are less likely to smoke.
- ▶ **Increase #of male students would not remove the bias.**

Causes of systematic errors

- Interviewer effect
- Respondents effect
- Knowing of the study purpose
- **Non-response**: It is the failure to obtain information on some of the subjects included in the sample to be studied
- Non response results in **significant bias when the following two conditions** are both fulfilled:

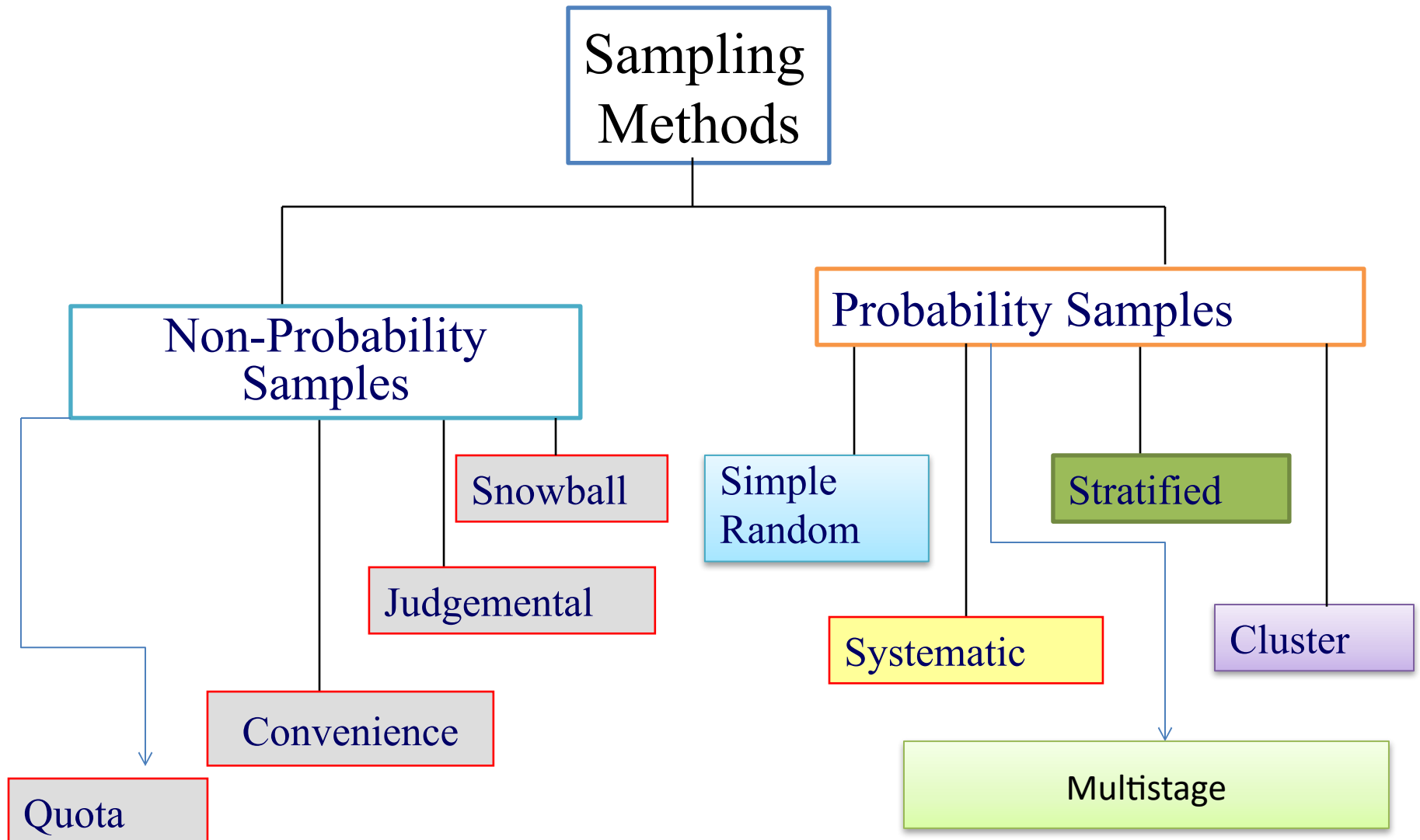
Causes of systematic errors...

- When non-response constitute significant portion of sample(15% or more)
- When non-respondent significantly differ from respondents
- **While selecting a SAMPLE, there are basic questions:**
 - What is the group of people (STUDY POPULATION) from which we want to draw a sample?
 - How many people do we need in our sample?
 - How will these people be selected?

Characteristics of a Good Sample Design

- ➡ Sample design must result in a truly **representative sample**.
- ➡ Sample design must be such which results in a **small sampling error**.
- ➡ Sample design must be viable in the context of **funds available for the research study**.
- ➡ Sample design must be such so that **systematic bias can be controlled in a better way**.
- ➡ Sample should be such that the results of the sample study can be applied, **in general, for the universe with a reasonable level of confidence**

Sampling Methods



A. Probability sampling

- Involves random selection of a sample
- Every sampling unit has a known and non-zero probability of selection into the sample.
- Involves the selection of a sample from a population, based on chance
- Can give as representative sample
- Best sampling methods if we want to generalize our findings to the population

Brain storming

Which sampling technique is appropriate for the following

1. When the population is homogenous with respect to the outcome variable and sample frame is available
2. When the population is homogenous with respect to the outcome variable and sample frame is not available
3. When the population is heterogeneous with respect to the outcome variable and sample frame is available
4. When the population is heterogeneous with respect to the outcome variable and sample frame is not available
5. When the study covers wide geographic area

1. Simple random sampling

- **Objective:** to select n units of population from N .
- Each member of a population has an equal chance of being included in the sample.
- **Assumptions of the population:**
 - Homogeneity with respect to the variable of interest
 - Availability of frame

Simple random sampling...

- **To use a SRS method:**
 1. Make a numbered list of all the units in the population
i.e. Sampling frame
 2. Each unit should be numbered from 1 to N (where N is the size of the population)
 3. Select the required number.

Simple random sampling...

- **The randomness of the sample is ensured by:**
 - Use of “lottery” methods
 - Table of random numbers
 - Computer programs

Simple random sampling...

Limitations of SRS

- Requires a sampling frame.
- Difficult if the reference population is dispersed.
- Minority sub-groups may be missed

2. Systematic random sampling

- Sometimes called *interval sampling*
- The population is assumed to be homogeneous, but no frame is available
- Selection of individuals from the sampling frame systematically
- Individuals are taken at regular intervals down the list
- The starting point is chosen at random

Systematic random sampling...

- Important if the reference population is arranged in some order:
 - Order of registration of patients
 - Numerical number of house numbers
 - Student's registration books
- Taking individuals at fixed intervals (every k^{th}) based on the sampling fraction

Systematic random sampling...

- Steps you need to follow in order to achieve a systematic random sample:
 1. Number the units in the population from 1 to N
 2. Decide on the n (sample size) that you want or need
 3. $k = N/n$ = the interval size
 4. Randomly select an integer between 1 to k
 5. Then take every k th unit

Note: *Systematic sampling should not be used when a cyclic repetition is inherent in the sampling frame.*

Systematic random sampling...

Example

- Select 20 samples from a total of 100 population using systematic random sampling technique?

3. Stratified random sampling _

- It is done when the population is known to have **heterogeneity** with regard to some factors and those factors are used for stratification
- Using **stratified sampling**, the population is divided into **homogeneous**, mutually exclusive groups called **strata**
- A population can be stratified by any variable that is available for all units prior to sampling (e.g., age, sex, province of residence, income, etc.).

Stratified random sampling ...

- A separate sample is taken independently from each stratum.
- Assumptions of the population:
 - Heterogeneity
 - If frame available, simple random will follow
 - If not, systematic will follow stratified sampling

Stratified random sampling

- Divide the population into **non-overlapping groups** (i.e., strata) $N_1, N_2, N_3, \dots, N_i$, such that $N_1 + N_2 + N_3 + \dots + N_i = N$.
- Elements within each strata are **homogeneous**, but are **heterogeneous across strata**.
- A simple random or a systematic sample is taken from each stratum relative to the proportion of that stratum to each of the others

Stratified random sampling...

- There are different sample allocation methods in order to select sample from each strata:
 1. **Equal allocation:** Allocate equal sample size to each stratum
 2. **Proportionate allocation:** allocating sampling proportional to the total population of each strata using the formula
- $$n_i = \frac{n}{N} * N_i$$

Where n=total sample size to be selected

n_i = is sample size from each stratum

N_j =Total population of the each stratum

N=total population

Advantages of Stratified sampling

- Every unit in a stratum has the same chance of being selected.
- Using the same sampling fraction for all strata ensures proportionate representation in the sample of the characteristic being stratified.
- Adequate representation of minority subgroups of interest can be ensured by stratification and
- By varying the sampling fraction between strata as required.

4. Cluster sampling

- A cluster sample is a simple random sample of groups or clusters of elements (vs. a simple random sample of individual objects)
 - Sometimes it is too expensive to carry out SRS
 - Population may be **large and scattered**.
 - Complete list of the study population unavailable
 - Travel costs can become **expensive** if interviewers have to survey people from one end of the country to the other.

Cluster sampling...

- A number of clusters are selected randomly to represent the total population, and then all units within selected clusters are included in the sample.
- This differs from stratified sampling, where some units are selected from each group.
- The clusters should be homogeneous, unlike stratified sampling where the strata are heterogeneous

Example

- In a school based study, we assume students of the same school are homogeneous.
- We can select randomly sections and include all students of the selected sections only

Cluster sampling...

Advantages

- Simple procedure
- No need of sampling frame
- Provides logistical advantage
 - Cost reduction

Dis advantages

- Creates a loss of efficiency when compared with SRS.
- you do not have total control over the final sample size.

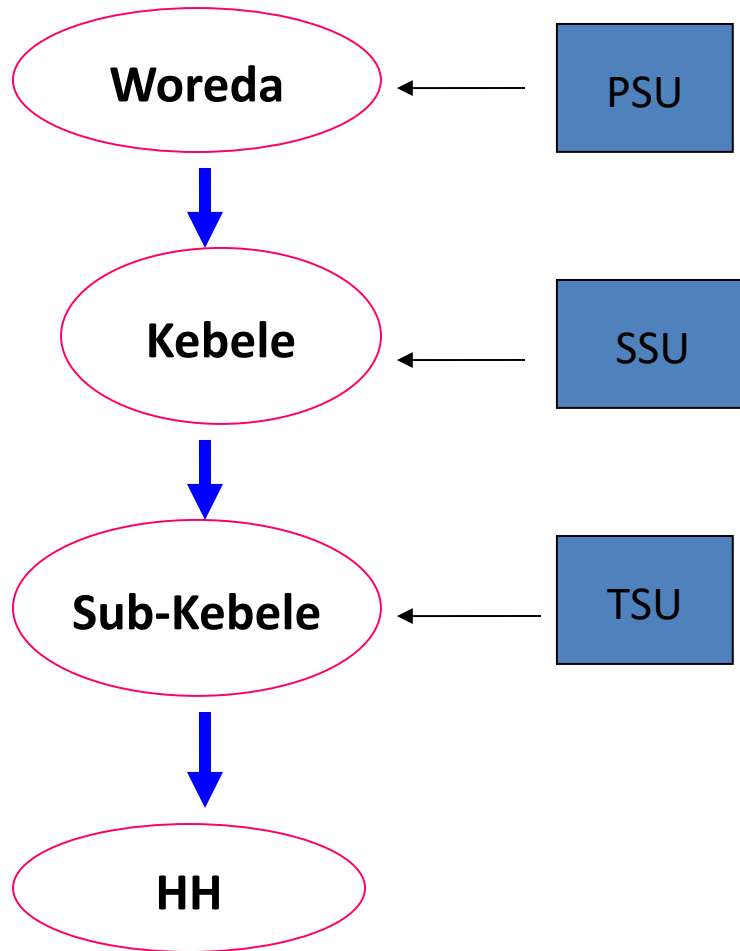
5. Multi-stage sampling

- Many surveys use complex sample designs that combine several of the above elements in a multi stage sampling framework
- Sometimes the population is **too large and scattered** for it to be practical to make a list of the entire population from which to draw a SRS.
- Suppose that each unit in the population can be divided into a number of smaller non-overlapping units, or subunits
- Similar to the cluster sampling, except that it **involves picking a sample from within each chosen cluster**

Multi-stage sampling...

- Most large surveys carried out this way
- The prime stimulus for multi-stage sampling is administrative convenience
- This type of sampling requires at least two stages.
- The *primary sampling unit (PSU)* is the sampling unit in the first sampling stage.
- The *secondary sampling unit (SSU)* is the sampling unit in the second sampling stage, etc.

Example:



Reading assignment

- Sequential sampling ?

B. Non-probability sampling

- In non-probability sampling, every item has an **unknown chance of being selected**.
- It does not involve random selection
- In non-probability sampling, there is an assumption that **there is an even distribution of a characteristic of interest within the population**.
- Non- probability sampling techniques are used when it is impossible to use probability sampling
they are *quick*, *inexpensive* and *convenient*

Reliability cannot be measured in non-probability sampling;
the only way to address data quality is to *compare some of the survey results with available information about the population.*
Estimation for the entire population is not possible
The accuracy of the survey results can not be determined

1. Convenience or haphazard sampling

- **Convenience sampling** is sometimes referred to as *haphazard* or *accidental sampling*.
- It is not normally representative of the target population because sample units are only selected if they can be accessed easily and conveniently.
- The obvious advantage is that the method is easy to use, but that advantage is greatly offset by the presence of bias.
- Although useful applications of the technique are limited, it can deliver accurate results when the population is homogeneous.

2. Volunteer sampling

- As the term implies, this type of sampling occurs when **people volunteer to be involved in the study.**
- In psychological experiments or pharmaceutical trials (drug testing), for example, it would be difficult and unethical to enlist random participants from the general public.
- In these instances, the sample is taken from a group of volunteers.
- Sometimes, the researcher offers payment to attract respondents.

In exchange, the volunteers accept the possibility of a lengthy, demanding or sometimes unpleasant process.

*Sampling voluntary participants as opposed to the general population may introduce **strong biases**.*

The silent majority does not typically respond, resulting in large **selection bias**.

3. Judgment sampling _

- This approach is used when a sample is *taken based on certain judgments about the overall population*.
- The underlying assumption is that the investigator will select units that are characteristic of the population.
- The critical issue here is **objectivity**: how much can judgment be relied upon to arrive at a typical sample?
- **Judgment sampling** is subject to the researcher's biases.
- One advantage of **judgment sampling** is the **reduced cost and time** involved in acquiring the sample.

4. Quota sampling

- ✓ It is the non probability equivalent of **stratified sampling**.
- ✓ Like stratified sampling, the researcher first identifies the **stratums** and their **proportions** as they are represented in the population.
- ✓ Then **convenience or judgment** sampling is used to select the **required number of subjects** from each stratum.
- ✓ This differs from stratified sampling, where the strata are filled by random sampling.

Quota sampling...

- **Quota sampling** is generally **less expensive** than random sampling.
- It is also **easy to administer**
- ***It*** is an effective sampling method when information is **urgently required and can be conducted without sampling frames.**
- In many cases where the population has no suitable frame, quota sampling may be the only appropriate sampling method.

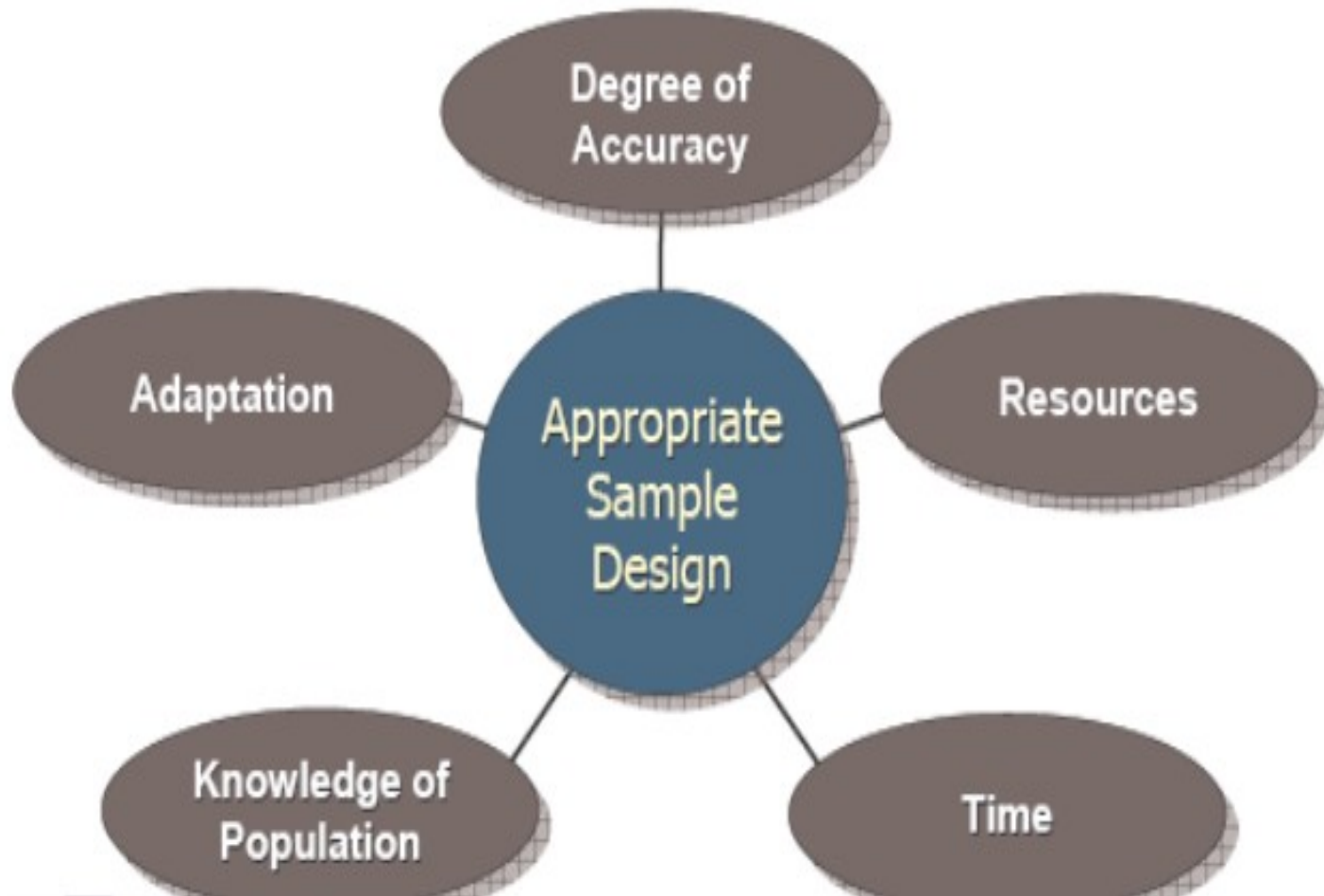
5. Snowball sampling

- A technique for selecting a research **sample** where existing study subjects recruit future subjects
- This sampling technique is often used **in hidden populations which are difficult for researchers to access**
- Example: populations would be drug users or commercial sex workers.
- Snowball samples are subject to numerous biases
 - For example, people who have many friends are more likely to be recruited into the sample.

Reading Assignment

- Types of snow ball sampling?
- Genealogy sampling ?

What is the appropriate sampling design?



Advantages of sampling

- **Save costs:** Less expensive to study the sample than the population.
- **Save time:** Less time needed to study the sample than the population.
- **Accuracy:** Since sampling is done with care and studies are conducted by skilled and qualified interviewers, the results are expected to be accurate

- **Sampling distribution**

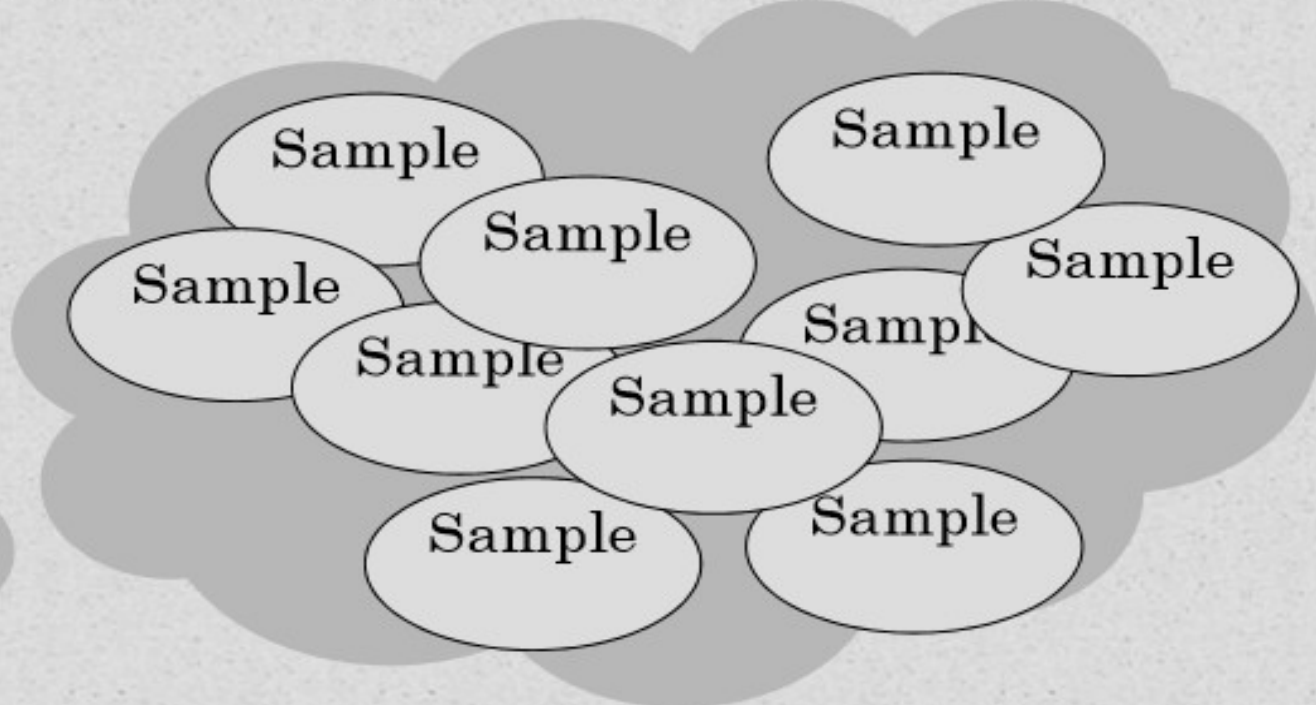
Objectives

- **At the end of this session student should able to:**
 - Discusses what mean sampling Distribution.
 - Differentiate main types of sampling distributions
 - Describe properties of sampling distribution of mean
 - Describe properties of sampling distribution of proportion

Sampling Distribution....

Sampling distribution can be defined as the probability distribution of a sample statistic that is formed when samples of size n are repeatedly taken from a population.

Population



Sampling distribution...

- Three things about sampling distribution of a statistic
 - its mean
 - its variance
 - its shape
- Due to random variation different samples from the same population will have different sample means.
- If we repeatedly take sample of the same size n from a population the means of the samples form a sampling distribution of means of size n is equal to population mean.

Sampling distribution...

Take a sample (n) from N and calculate the statistic, e.g., mean.

- Take another sample (same size) and calculate mean.
- Repeat & repeat & repeat &
- Put all these sample statistics together to get a distribution of sample statistics (Frequency distribution).

Sampling distribution...

Main types of sampling distributions

- A. Distribution of the sample mean
- B. Distribution of the difference between two means
- C. Distribution of the sample proportion
- D. Distribution of the difference between two proportions

A. Sampling distribution of sample mean

- Suppose we have a population of size **N=4**, constituting the ages of four outpatients and we are going to collect **n=2** samples repeatedly .

x, Age (years): 18, 20, 22, 24

$$\begin{aligned}\mu &= \frac{\sum x_i}{N} \\ &= \frac{18 + 20 + 22 + 24}{4} = 21\end{aligned}$$

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}} = 2.236$$

Sampling distribution...

1 st	2 nd Observation			
Obs	18	20	22	24
18	18,18	18,20	18,22	18,24
20	20,18	20,20	20,22	20,24
22	22,18	22,20	22,22	22,24
24	24,18	24,20	24,22	24,24



1 st	2 nd Observation			
Obs	18	20	22	24
18	18	19	20	21
20	19	20	21	22
22	20	21	22	23
24	21	22	23	24

- 16 possible samples (with replacement)

- 16 Sample Means

Sampling distribution...

<i>Sample means</i>	<i>Freq</i>	<i>$P(\bar{x})$</i>
18	1	0.0625
19	2	0.1250
20	3	0.1875
21	4	0.2500
22	3	0.1875
23	2	0.1250
24	1	0.0625

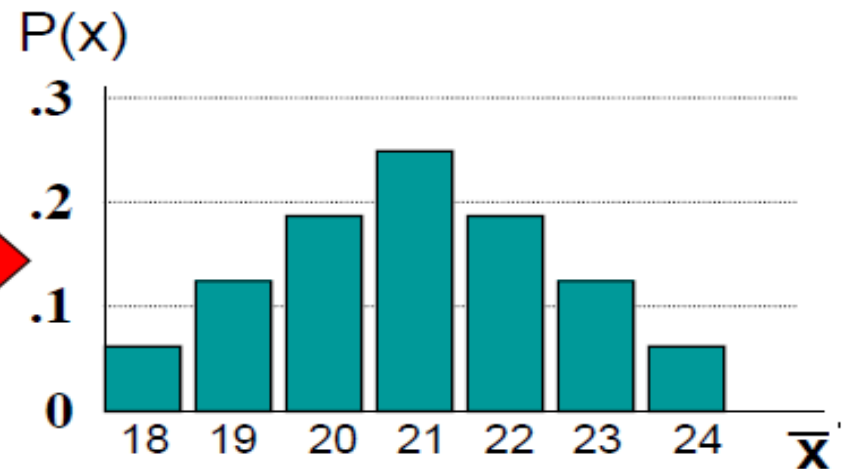
Sampling distribution...

Sampling distribution of all sample means

16 Sample Means

1st Obs	2nd Observation			
	18	20	22	24
18	18	19	20	21
20	19	20	21	22
22	20	21	22	23
24	21	22	23	24

Sample Means Distribution



Sampling distribution...

Summary measures of this sampling distribution:
Add the 16 sample means & divide by 16. Also calculate the SD of the sample means.

$$\mu_{\bar{x}} = \frac{\sum \bar{x}_i}{N} = \frac{18 + 19 + 21 + \dots + 24}{16} = 21$$

$$\begin{aligned}\sigma_{\bar{x}} &= \sqrt{\frac{\sum (x_i - \mu_{\bar{x}})^2}{N}} \\ &= \sqrt{\frac{(18 - 21)^2 + (19 - 21)^2 + \dots + (24 - 21)^2}{16}} = 1.58\end{aligned}$$

Properties of Sampling Distribution.....

Sampling Distribution of the mean: Suppose we choose a random sample of size n , the sampling distribution of the sample mean $\bar{x}$ possesses the following properties.

- ▣ The sample mean $\bar{x}$ will be an estimate of the population mean μ .
- ▣ The standard deviation of $\bar{x}$ is $\sigma/\sqrt{n}$ (called the standard error of the mean).
- ▣ Provided n is large enough the shape of the sampling distribution of $\bar{x}$ is normal.

B. Sampling Distribution of Proportion

□ Proportion

- Suppose we choose a random sample of size n , the sampling distribution of the sample proportion p possesses the following properties.
- The sample proportion p will be an estimate of the population mean p .
- The standard deviation of p is $\sqrt{p(1-p)/n}$ (called the standard error of the proportion).
- Provided n is large enough the shape of the sampling distribution of p is normal.

The Central Limit Theorem

for non-normally distributed data

- Regardless of the shape of the frequency distribution of a characteristic in the parent population,
 - the means of a large number of samples (independent observations) from the population will follow a **normal distribution** (with the mean of means approaches the population mean μ , and standard deviation of $\sigma/\sqrt{n}$).

Central limit theorem...

- Inferential statistical techniques have various assumptions that must be met before valid conclusions can be obtained
 - Samples must be randomly selected.
 - sample size must be greater ($n \geq 30$) or
 - the population must be normally or approximately normally distributed if the sample size is less than 30.
- NB: Generally, how much large sample depends on the shape of the distribution of the sampled population
 - The greater the skewness, the larger the sample size it need

Standard deviation and Standard error

- ▶ **Standard deviation** is a measure of variability between individual observations (descriptive index relevant to mean)
- ▶ **Standard error** refers to the variability of summary statistics (e.g. the variability of the sample mean or a sample proportion)
- ▶ SE is determined by both the sample size and the degree of variability among the individual observations
- ▶ Standard error is a measure of uncertainty in a sample statistics i.e. precision of the estimate of the estimator
- ▶ Always SE is less than standard deviation

Applications of the sampling distributions of sample mean

- ✚ Helps in computing the probability of obtaining a sample with a mean of some specified magnitude.

Applications of the sampling mean ...

z-value for sampling distribution of $\bar{x}$

$$z = \frac{(\bar{x} - \mu)}{\frac{\sigma}{\sqrt{n}}}$$

where:

$\bar{x}$ = sample mean

μ = population mean

σ = population standard deviation

n = sample size

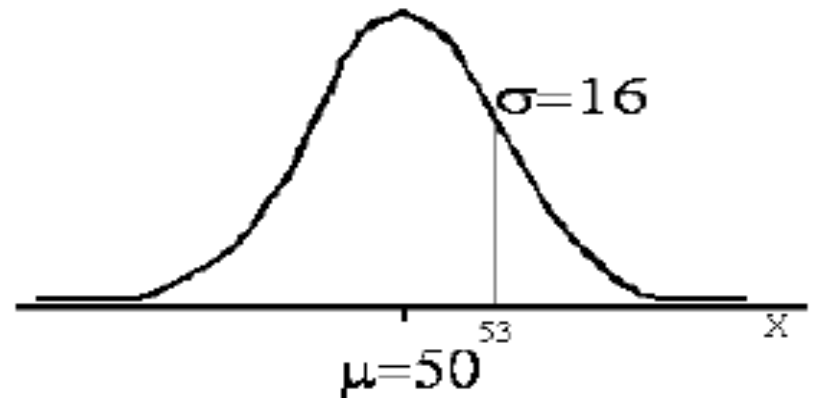
Example 1

- **Given: $\mu = 50$, $\sigma = 16$, $n = 64$**

Find: $P(x > 53)$

Solution

1. Write the given information, $\mu=50$, $\sigma=16$, $n=64$
2. Sketch a normal curve



Example 1 cont...

3. Convert x to a z score

$$z = \frac{x - \mu}{\sigma / \sqrt{n}}$$
$$z = \frac{53 - 50}{16 / \sqrt{64}} = \frac{3}{2} = 1.5$$

4. Find the appropriate value(s) in the Table

The area of the SND above a value of $z = 1.5$ gives an area of 0.0668. The probability $P(z > 1.5) = 0.0668$

5. Complete the answer

The probability that X is greater than 53 is 0.0668.

Example 2

- Suppose a population has mean $\mu = 8$ and standard deviation $\sigma = 3$. **Suppose a random sample of size $n = 36$ is selected.**
- What is the probability that the sample mean is between 7.8 and 8.2?
 - ... with mean $\mu_{\bar{x}} = 8$
 - ...and $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{3}{\sqrt{36}} = 0.5$

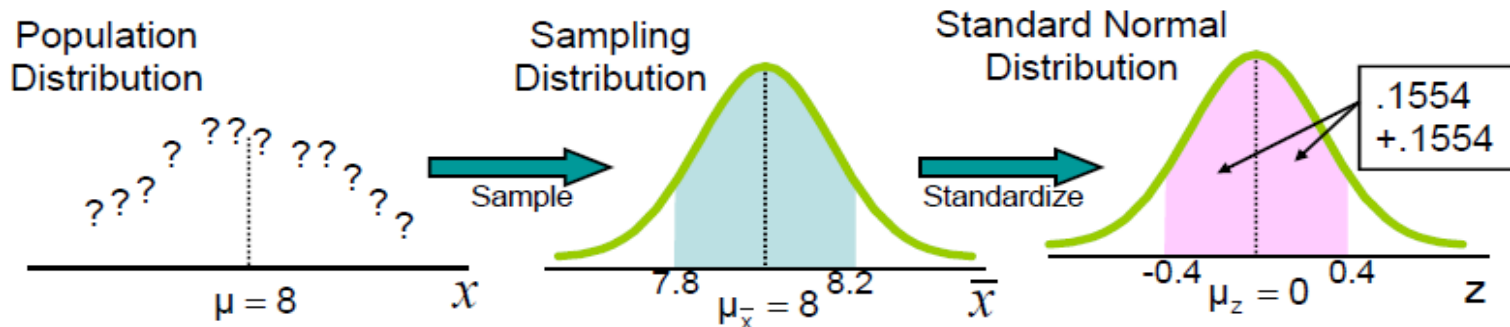
Example 2 cont...

Solution:

- Even if the population is not normally distributed, the central limit theorem can be used (**$n > 30$**)
- ... so the sampling distribution of $\bar{x}$ is approximately normal

Example 2 cont...

$$\begin{aligned} P(7.8 < \mu_{\bar{x}} < 8.2) &= P\left(\frac{7.8 - 8}{3/\sqrt{36}} < \frac{\mu_{\bar{x}} - \mu}{\sigma/\sqrt{n}} < \frac{8.2 - 8}{3/\sqrt{36}}\right) \\ &= P(-0.4 < z < 0.4) = 0.3108 \end{aligned}$$



Example 3

- The distribution of serum cholesterol levels for all 20-70 year-old males has mean $\mu = 211$ mg/100 ml and $\sigma = 46$ mg/100 ml.
 - a. If a sample of size 25 is selected from this population, what is the probability that the sample has a mean of 230 or above?
- Since $\bar{x}$ has a normal distribution with mean 211 and standard error 9.2,

$$z = \frac{\bar{x} - 211}{9.2}$$

Example 3 cont...

- The area under the standard normal curve to the right of $z = 2.07$ is 0.0192
- Consequently, the probability that a sample of size 25 has a mean of 230 mg/100 ml or higher is **0.0192**.

Example 3 cont...

- b. What mean value of serum cholesterol level cuts off the lower 10% of the sampling distribution?
- An area of 0.1003 in the lower tail of the SND is marked by the value $z = -1.28$
 - What is the corresponding value of $\bar{x}$?
$$z = \frac{\bar{x} - 211}{9.2} = -1.28$$

Example 3 cont...

$$\begin{aligned}\bar{x} &= 211 + (9.2)(-1.28) \\ &= 199.2\end{aligned}$$

- Approximately 10% of samples of size 25 have means that are less than or equal to 199.2 mg/100 ml.
- The other 90% of the samples have means that are greater than 199.2 mg/100 ml

Exercise 1

- The distribution of serum cholesterol levels (normally distributed) for all 20-70 year-old males has mean $\mu = 211$ mg/100 ml and $\sigma = 64$ mg/100 ml. If a sample of size 36 is selected from this population :
 - A. what is the probability that the sample has a mean of 230 or above?
 - B. What is the probability that the sample mean is between 200 and 240?
 - C. What mean value of serum cholesterol level cuts off the upper 20% of the sampling distribution?

- **ANY QUESTION ???**